

### 5.1.1 Sequences

#### Definition

A **sequence** is a function whose domain is either all the integers between two given integers or all the integers greater than or equal to a given integer.

We typically represent a sequence as a set of elements written in a row. In the sequence denoted

$$a_m, a_{m+1}, a_{m+2}, \dots, a_n,$$

each individual element  $a_k$  (read “a sub k”) is called a **term**. The  $k$  in  $a_k$  is called a **subscript** or **index**,  $m$  (which may be any integer) is the subscript of the **initial term**, and  $n$  (which must be an integer that is greater than or equal to  $m$ ) is the subscript of the **final term**. The notation

$$a_m, a_{m+1}, a_{m+2}, \dots$$

denotes an **infinite sequence**. An **explicit formula** or **general formula** for a sequence is a rule that shows how the values of  $a_k$  depend on  $k$ .